



**UG Program in Electronics and Communication
(Advanced Communication Technology)**

Experiment No. 2

Computing entropy of data from the given probability distribution.

2.1 Aim: Write a python program for computing entropy of data from the given probability distribution.

2.2 Course Outcome: Apply the concepts of information theory in source coding.

2.3 Learning Objectives: Students will be able

Understand the concept of entropy in information theory: The objective is to understand how entropy quantifies uncertainty or randomness in data.

Apply entropy calculation to real-world datasets: Students can compute entropy for different types of probability distributions and interpret the result as the uncertainty of information.

2.5 Program and Output:

```
[3] ✓ Os #Heet Sedani - 125BTAC2005
from math import log2
p1=0.7
p2=0.5
H=p1*log2(1/p1)+p2*log2(1/p2)
print("Entropy:",H)

... Entropy: 0.8602012209808307
```



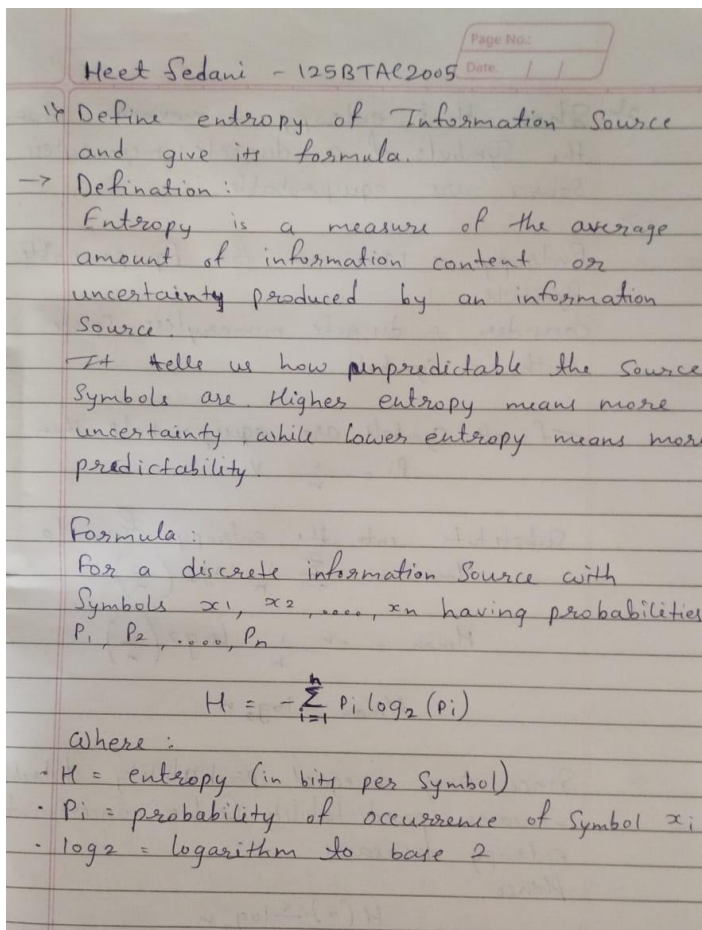
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2.6 Conclusion:

In this experiment, the entropy of data was computed using a given probability distribution with the help of a Python program. The results helped in understanding how entropy measures the amount of uncertainty or randomness present in an information source. It was observed that higher entropy indicates more uncertainty in data, while lower entropy represents more predictable information. Thus, this experiment successfully demonstrated the practical application of information theory concepts used in source coding and digital communication systems.

2.7 Questions:

1. Define entropy of Information source and give its formula





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2. Show that entropy is maximum when all the symbols of a discrete memoryless source are equiprobable.

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→ Entropy is maximum for Equiprobable Symbols

Consider a discrete memoryless source with 'n' symbols

If all symbols are equiprobable then

$$P_i = \frac{1}{n}, \forall i$$

Substitute into the entropy formula

$$H_{\max} = - \sum_{i=1}^n \frac{1}{n} \log_2 \left(\frac{1}{n} \right)$$
$$H_{\max} = -n \cdot \frac{1}{n} \cdot \log_2 \left(\frac{1}{n} \right)$$
$$H_{\max} = \log_2 n$$

Since any unequal probability distribution increases predictability (reduces uncertainty) entropy decreases

Hence,

$$H(x) \leq \log_2 n$$